Design and Manufacture
Research and specification pro forma

Design brief 3

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Focus group

A focus group of parents was formed to gather their thoughts on playground equipment. The parents were asked to discuss what their children would like in playground equipment and what they as parents thought it should include.

The main points from the discussion are summarised below:

- *All parents agreed that the equipment had to be easy to access and use, to allow children to feel as included as possible.*
- *When asked if there was anything we could consider, parents thought it was important that there were both vertical and horizontal surfaces to play at, as some children struggled with reaching up for long periods of time. It was agreed that this should be included in some way.*

Site visits

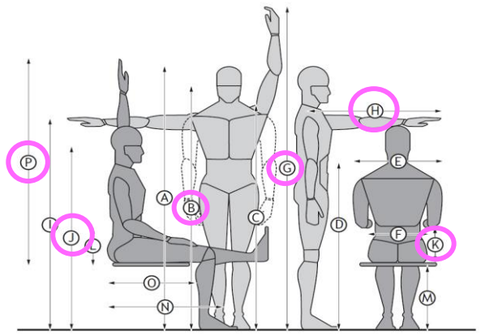
I visited playgrounds at a number of different holiday parks. This helped to identify the range of surfaces that the playground equipment would need to be secured to.





From the site visits, we identified that the three surface types used in the playgrounds were sand, grass and rubber matting.

Anthropometric data

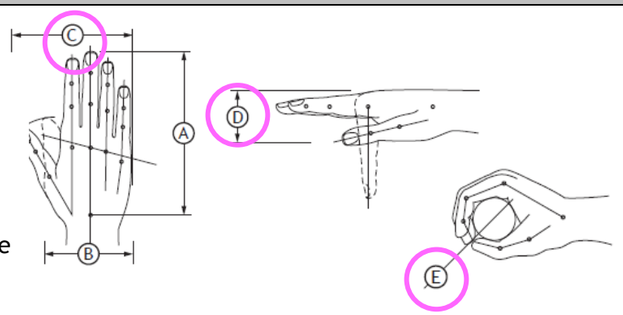


Standing and seated figures		4-year-old child		
		Percentile		
		5th	50th	95th
B	Eye level	872	951	1030
G	Vertical reach	1094	1211	1328
H	Forward reach	389	450	505
J	Eye level (seated)	435	482	490
K	Elbow height (seated)	122	138	150
P	Sitting reach	694	764	818

I have selected the 5th percentile for vertical reach, forward reach and sitting reach to ensure that all children could access parts of the playground equipment.

The 50th percentile was selected for eye level (both standing and seating) to ensure the sizes would meet the needs of the majority of children.

There would need to be enough clearance at elbow height for all children to move around the playground equipment, so I have used the 95th percentile for this.



Hand		4-year-old child		
		Percentile		
		5th	50th	95th
C	Hand span	58	66	73
D	Hand thickness	15	18	21
E	Grip diameter INDEX finger	62	75	87

I have chosen the 95th percentile hand span, thickness and grip diameter for 4-year-old children. I will need to add additional clearance to these to ensure it will also be suitable for 6-year-olds.

Specification

The specification should include key points from the brief and research to detail the requirements of the proposal.

It should clarify aspects such as aesthetics, function, ergonomics, performance, and cost.

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